

CE 1201

ENGINEERING MECHANICS

SECTION A - COMPLETE CLASS NOTE

Concept • Bangla Explanation • Shortcuts • Derivations • Sketches • QB Match

Set A1: Centroid & Center of Gravity

Set A2: Moment of Inertia of Area

Set A3: Product of Inertia • Mass MOI • Pappus-Guldinus

Set A4: Dynamics • Work-Energy • Impulse-Momentum

Prepared from the provided class notes, lecture slides and regular question bank

Made by 2223033

How to Use This Note

Exam structure: প্রতি Section-এ ৪টি set-এর মধ্যে ৩টি answer করতে হবে। তাই ৪টি set-এর সবগুলো একভাবে পড়ার বদলে ৩টি strong set + ১টি backup set তৈরি করা সবচেয়ে নিরাপদ।

এই নোটের “Set A1-A4” নামকরণ আপনার দেওয়া Section-A set-wise note structure অনুসরণ করে। বিভিন্ন বছরের printed question paper-এ topic কখনও অন্য section-এ ছাপা হয়েছে; এখানে QB matching topic-wise করা হয়েছে।

Set	Core topic	What to master first	Exam use
A1	Centroid & C.G.	Definitions c ommon shapes → composite area → sector/arc	Very high repeat
A2	Area MOI	$I_x/I_y/J$ k s t andard formulas parallel axis □ composite	Very high repeat
A3	Product/Mass MOI/Pappus	P_{xy} + principal I → mass formulas Pappus theorems	High-value mixed set
A4	Dynamics	$F=ma$ p rojectile □ work-energy → impulse-momentum	High repeat; numerical heavy

Best reading order: A1 → A2 → A4 → A3. কারণ A1/A2-এর geometry sense A3-কে সহজ করে, আর A4-তে formula selection practice বেশি দরকার।

Notation You Will See

Symbol	Meaning
$\bar{x}, \bar{y}$	Centroid coordinates
A	Area
I_x, I_y	Rectangular/area moments of inertia
J	Polar moment of inertia
k	Radius of gyration
P_{xy}	Product of inertia
m, W	Mass and weight
U or W_{work}	Work
T or KE	Kinetic energy
J_{imp}	Impulse
v, a, s, t	Velocity, acceleration, displacement, time

SET A1 - Centroid, Center of Gravity & Center of Mass

Memory line: Centroid = Geometry-এর balance point; Center of Mass = mass distribution-এর balance point; Center of Gravity = weight/gravity-এর resultant action point। Uniform density + uniform gravity হলে এগুলো একই point-এ coincide করতে পারে।

1.1 Centroid - Basic Idea

Centroid হলো একটি line/area/volume-এর geometric center বা প্রথম moment balance হওয়ার point। Uniform lamina বা uniform-density geometric body-র ক্ষেত্রে centroid-কে সহজভাবে “shape-এর ভারসাম্য বিন্দু” হিসেবে ভাবা যায়।

- Area centroid-এর জন্য: $\bar{x} = (x \, dA)/A$, $\bar{y} = (y \, dA)/A$
- Composite area-এর জন্য: $\bar{x} = \Sigma(A_i x_i)/\Sigma A_i$, $\bar{y} = \Sigma(A_i y_i)/\Sigma A_i$
- Hole/cutout হলে area negative ধরা হয়।

$$\bar{x} = \Sigma(A_i x_i) / \Sigma A_i \quad ; \quad \bar{y} = \Sigma(A_i y_i) / \Sigma A_i$$

QB MATCH Matching Question Bank - Centroid definition & applications

Year(s)	Matching question / pattern	Priority
2015-2021	Define centroid; real-life applications; locate centroid of standard/composite area.	★★★★★
2023	Distinguish centroid and center of gravity; real-life applications in building engineering.	★★★★★
2024	Centroid of area enclosed by parabolas + composite unshaded area.	★★★★☆
2025	Condition for C.M./C.G. coincidence; sector centroid + composite C.G.	★★★★★

1.2 Centroid vs Center of Mass vs Center of Gravity

Term	What it depends on	Best Bangla meaning	Key relation
Centroid	Geometry only	জ্যামিতিক কেন্দ্র	Uniform density হলে mass effect cancels
Center of Mass (C.M.)	Mass distribution	ভরের কেন্দ্র	$\bar{x} = \int x \, dm / m$
Center of Gravity (C.G.)	Weight + gravity field	ওজনের কার্যকর কেন্দ্র	$\bar{x} = \int x \, dW / W$

$$\text{C.G.: } \bar{x} = \int x \, dW / W \quad \text{C.M.: } \bar{x} = \int x \, dm / m$$

Exam-safe statement: Gravity field uniform and parallel হলে center of gravity center of mass-এর সঙ্গে coincide করে। Body homogeneous হলে geometric centroid-ও একই point-এ হতে পারে।

QB MATCH Matching Question Bank - C.G./C.M.

Year(s)	Matching question / pattern	Priority
2019, 2023	Centroid and center of gravity: meaning/difference/application.	★★★★☆
2025	Explain condition under which center of mass coincides with center of gravity.	★★★★★
Class slides	Derive C.G. coordinates from principle of moments; show C.M. equations.	Theory

1.3 First Moment of Area - Why the Formula Works

কোনো axis-এর প্রতি area-এর first moment হলো area $\times$ perpendicular distance-এর algebraic accumulation। Centroid এমন point যেখানে centroidal axis-এর প্রতি total first moment balance করে।

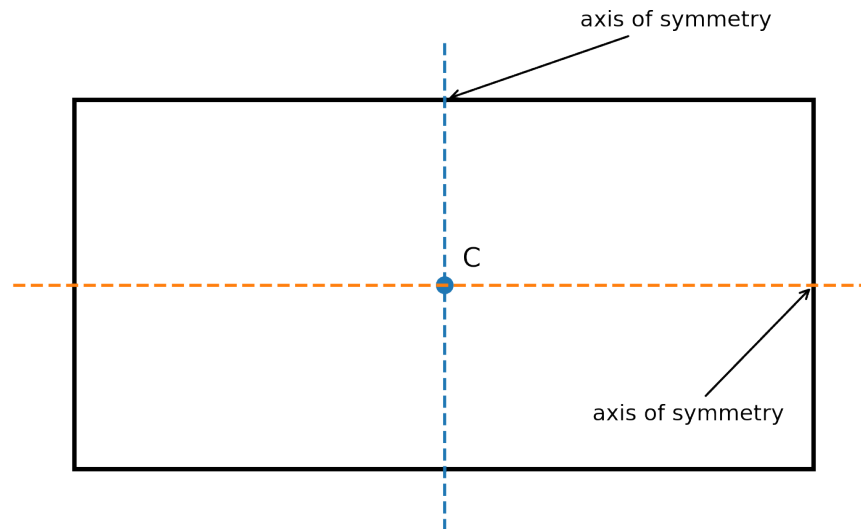
$$Q_x = \int y \, dA = A\bar{y} \quad ; \quad Q_y = \int x \, dA = A\bar{x}$$

Shortcut: Centroid problem দেখলেই “Area $\times$ distance” ভাবুন। Moment of Inertia problem দেখলেই “Area $\times$ distance²” ভাবুন। এই একটি distinction অনেক confusion কমায়।

1.4 Axis of Reference & Principle of Symmetry

Reference axis হলো যে axis থেকে $\bar{x}$ এবং $\bar{y}$ -এর distance মাপা হয়। সাধারণত figure-এর left-most vertical line-কে y-axis এবং bottom-most horizontal line-কে x-axis ধরা সুবিধাজনক।

Principle of symmetry: কোনো plane area-এর একটি symmetry axis থাকলে centroid ওই axis-এর উপর থাকবে; দুটি symmetry axis থাকলে তাদের intersection-এ centroid থাকবে।



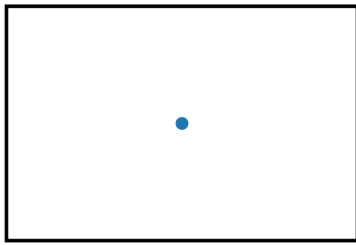
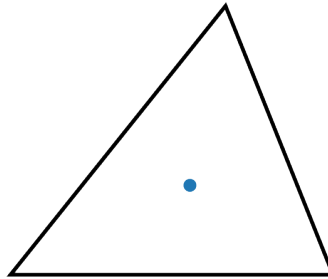
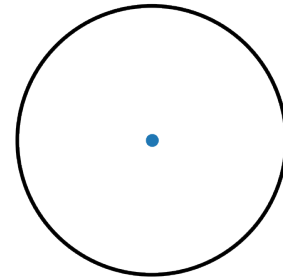
Symmetry থাকলে unknown coordinate দ্রুত নির্ণয় করা যায়।

মনে রাখার কৌশল: “Symmetry saves calculation.” আগে symmetry খুঁজুন, পরে formula ব্যবহার করুন।

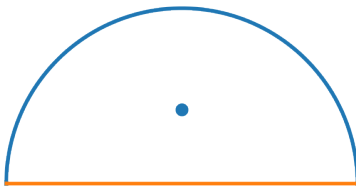
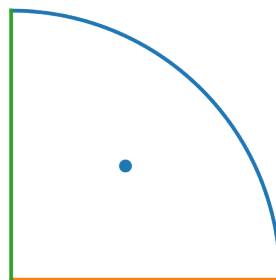
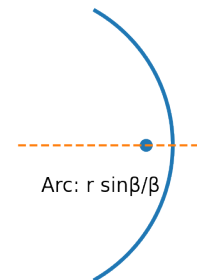
QB MATCH Matching Question Bank - Axis of symmetry

Year(s)	Matching question / pattern	Priority
2017, 2018	Describe/illustrate axis of symmetry with centroid implication.	★★★★★
2019	Centroid/C.G. + sector and symmetry-related geometric reasoning.	★★★★☆

1.5 Centroid of Common Shapes - Must Memorize

Rectangle: $(b/2, h/2)$ Triangle: $h/3$ from base

Circle: center

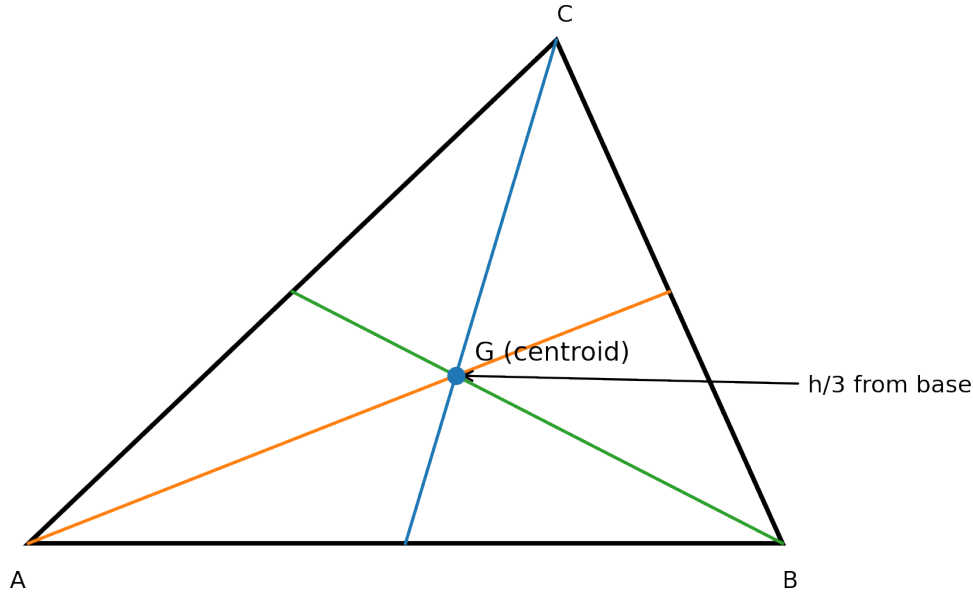
Semicircle: $4r/(3\pi)$ Sector: $2r \sin\beta/(3\beta)$ Arc: $r \sin\beta/\beta$

β must be in radians for sector/arc formulas.

Shape	Centroid location	Memory cue
Rectangle $b \times h$	$(b/2, h/2)$	Half-half
Triangle	$h/3$ from base = $2h/3$ from opposite vertex	Base $\square 1/3$
Circle	Center	Full symmetry
Semicircular area	$4r/(3\pi)$ from diameter along symmetry axis	Area $\pi r^2/2$
Quarter-circular area	$\bar{x} = \bar{y} = 4r/(3\pi)$	Same in x and y
Circular sector, angle 2β	distance = $2r \sin\beta/(3\beta)$ from center	Sector has 2/3 factor
Circular arc, angle 2β	distance = $r \sin\beta/\beta$ from center	Arc has no 2/3
Semicircular arc	$2r/\pi$ from center	Line/arc result

Sector vs Arc: Sector = area; Arc = line. তাই sector formula-তে $2/3$ factor আছে, arc formula-তে নেই।

1.6 Derivation - Centroid of a Triangle



For a triangle, the centroid lies at the intersection of medians.

Base b এবং height h -এর triangle ধরা হলো। Base থেকে y distance-এ একটি thin horizontal strip নেওয়া হলে similar triangle থেকে strip width $= b(h-y)/h$ ।

$$dA = [b(h - y)/h] dy$$

তারপর $\bar{y} = (\int y dA) / (\int dA)$, limit $y = 0$ থেকে h ।

$$\bar{y} = \{ \int_0^h y \cdot [b(h-y)/h] dy \} / (bh/2) = h/3$$

Final: Triangle centroid = base থেকে $h/3$; vertex থেকে $2h/3$ । Median-কে centroid 2:1 ratio-তে ভাগ করে।

QB MATCH Matching Question Bank - Triangle centroid derivation/result

Year(s)	Matching question / pattern	Priority
2016, 2018, 2021	Find centroid of triangle of base b and altitude h .	★★★★★
Theory note	Use integration/similar triangles to obtain $h/3$.	Derivation

1.7 Circular Sector & Circular Arc

Central angle $= 2\beta$ হলে centroid symmetry axis-এর উপর থাকবে। β radians-এ নিতে হবে।

$$\text{Sector area: } \bar{x} = 2r \sin\beta / (3\beta)$$

$$\text{Circular arc: } \bar{x} = r \sin\beta / \beta$$

QB MATCH Matching Question Bank - Sector / Arc

Year(s)	Matching question / pattern	Priority
2017, 2019	Locate centroid of a circular sector of radius r subtending 2β .	★★★★★
2023	Locate centroid of sector of circle radius r , central angle 2β .	★★★★★
2025	Determine centroid of a sector of circle radius r , angle 2β .	★★★★★
Class note	Locate centroid of circular arc subtending 2β .	High-value theory

1.8 Composite Area Method - Universal Procedure

- Figure-কে basic shapes-এ ভাগ করুন: rectangle, triangle, circle, semicircle, sector ইত্যাদি।
- একটি convenient x-y reference axis নিন।
- প্রতিটি অংশের area A_i লিখুন; hole/cutout হলে negative।
- প্রতিটি অংশের centroid coordinate x_i, y_i বের করুন।
- $A_i x_i$ এবং $A_i y_i$ হিসাব করুন।
- $\bar{x} = \Sigma A_i x_i / \Sigma A_i$ এবং $\bar{y} = \Sigma A_i y_i / \Sigma A_i$
- Final coordinate figure-এর physical boundary-এর সঙ্গে sanity-check করুন।

Part	A_i	x_i	y_i	$A_i x_i$	$A_i y_i$
1	$+A_1$	x_1	y_1	$A_1 x_1$	$A_1 y_1$
2	$+A_2$	x_2	y_2	$A_2 x_2$	$A_2 y_2$
Hole	$-A_3$	x_3	y_3	$-A_3 x_3$	$-A_3 y_3$
Σ	ΣA			$\Sigma A x$	$\Sigma A y$

Error trap: Hole-এর area negative করলেও x_i, y_i distance নিজে negative হয় না যদি centroid positive quadrant-এ থাকে; sign মূলত A_i -তে বসান।

QB MATCH Matching Question Bank - Composite centroid

Year(s)	Matching question / pattern	Priority
2015-2021	Almost every year: compute $\bar{x}, \bar{y}$ of shaded/unshaded composite area.	★★★★★
2023	Find centroid about x and y axes of unshaded area with circular cutouts.	★★★★★
2024	Composite unshaded area + area between parabolic arcs.	★★★★★
2025	Locate C.G. of composite area with curved cutouts.	★★★★★

1.9 Special Curves - Parabola-Type Question

Parabolic boundary থাকলে standard centroid table available না থাকলে integration ব্যবহার করতে হবে। Area enclosed by curves-এর জন্য আগে intersection limits বের করুন, তারপর dA নির্বাচন করে A , $\int x dA$, $\int y dA$ হিসাব করুন।

$$A = \int dA, \quad \bar{x} = (1/A) \int x dA, \quad \bar{y} = (1/A) \int y dA$$

Technique: Curves $y^2 = kx$ ও $x^2 = ky$ ধরনের হলে symmetry পরীক্ষা করুন। x - y interchange করলে shape symmetric কিনা দেখলে অনেক সময় $\bar{x} = \bar{y}$ সরাসরি পাওয়া যায়।

QB MATCH Matching Question Bank - Parabolic centroid

Year(s)	Matching question / pattern	Priority
2022 note pattern	Centroid of area enclosed by $y^2 = 4x$ and $x^2 = 4y$.	★★★★☆
2024	Centroid of area enclosed by $y^2 = 9x$ and $x^2 = 4y$.	★★★★★

1.10 A1 Exam Checklist

Before exam	Must be able to do
Theory	Centroid, C.G., C.M., axis of symmetry, applications, condition of coincidence
Formulas	Triangle, rectangle, semicircle, quarter circle, sector, arc
Math	Composite area with holes; one curved/parabolic case
Derivation	Triangle centroid; sector/arc result explanation

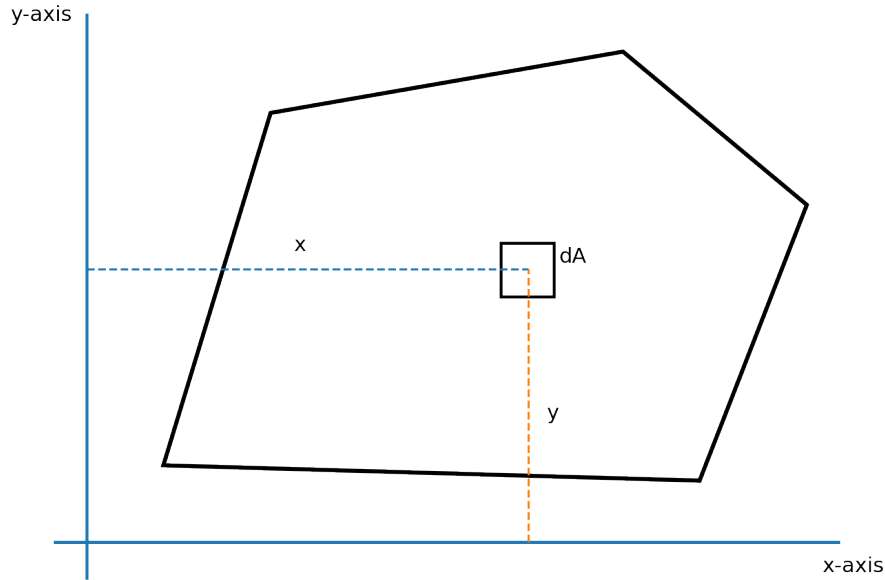
A1 one-line mnemonic: "Shape → Split → Sign → Centroid → ΣA → $\Sigma Ax/\Sigma Ay$."

SET A2 - Moment of Inertia of Area

Core idea: Centroid-এ distance ছিল first power; Area MOI-তে distance squared। দূরে থাকা area element বেশি contribution দেয়।

2.1 What is Area Moment of Inertia?

Second moment of area হলো একটি purely geometric property, যা area কীভাবে একটি axis থেকে ছড়িয়ে আছে তা measure করে। Beam bending/deflection ও stress analysis-এ এটি খুব গুরুত্বপূর্ণ।



dA element at distances x and y from the axes.

$$I_x = \int y^2 dA \quad ; \quad I_y = \int x^2 dA$$

Unit: Area MOI-এর unit length⁴: mm⁴, cm⁴, m⁴, in⁴, ft⁴। Mass MOI-এর unit আলাদা - এটি Set A3-তে দেখানো হয়েছে।

QB MATCH Matching Question Bank - Definition / significance

Year(s)	Matching question / pattern	Priority
2015-2021	Define rectangular/second moment, polar MOI; applications/significance.	★★★★★
2023	Define moment of inertia and product of inertia; triangle MOI + radius of gyration.	★★★★★
2024-2025	Significance/basic concept + composite centroidal MOI.	★★★★★

2.2 Rectangular MOI, Polar MOI and Perpendicular-Axis Relation

Plane area-এর জন্য z-axis যদি area-এর plane-এর perpendicular হয় এবং x-y axes একই point-এ intersect করে, তবে polar moment J সেই point-এর about measure করা হয়।

$$J = \int r^2 dA = \int (x^2 + y^2) dA = I_x + I_y$$

Proof shortcut: $r^2 = x^2 + y^2$ লিখুন $\rightarrow$ integral split করুন $\rightarrow \int x^2 dA = I_y, \int y^2 dA = I_x$ ।

QB MATCH Matching Question Bank - Prove $J = I_x + I_y$

Year(s)	Matching question / pattern	Priority
2017, 2018, 2019, 2021	Define rectangular/polar MOI and prove $J = I_x + I_y$.	★★★★★
2023-2025	Use polar MOI in composite cross-section / radius of gyration problem.	★★★★★

2.3 Radius of Gyration

Radius of gyration k হলো এমন একটি equivalent distance - যদি পুরো area A ওই distance-এ concentrate করা হয়, তবে একই MOI পাওয়া যাবে।

$$k_x = \sqrt{I_x/A}, \quad k_y = \sqrt{I_y/A}, \quad k_p = \sqrt{J/A}$$

$$I = Ak^2 \quad ; \quad J = Ak_p^2$$

Bangla intuition: “একই inertia পেতে area-টা axis থেকে কত দূরে এক জায়গায় বসাতে হবে?” - সেই distance-ই k।

2.4 Standard Area MOI Formulas

Shape / axis	Formula	Remember
Rectangle $b \times h$, centroidal x	$I_x = bh^3/12$	x-axis-এর perpendicular dimension h is cubed
Rectangle $b \times h$, centroidal y	$I_y = hb^3/12$	y-axis-এর perpendicular dimension b is cubed
Rectangle about base	$I = bh^3/3$	Centroidal $\times 4$
Triangle about base	$I = bh^3/12$	Base axis
Triangle centroidal axis $\parallel$ base	$\bar{I} = bh^3/36$	Base result /3
Circle about centroidal diameter	$I_x = I_y = \pi r^4/4$	Two equal diameters
Circle polar about center	$J = \pi r^4/2$	$= I_x + I_y$

Rectangle shortcut: যে axis-এর প্রতি I বের করবেন, axis-এর perpendicular dimension-টাই cube হবে। I_x -এ h^3 , I_y -এ b^3 ।

2.5 Derivation - Rectangle about Centroidal x-axis

Centroidal x-axis থেকে y distance-এ width b এবং thickness dy-এর strip নিন। $dA = b dy$ ।

$$I_x = \int y^2 dA = b \int_{-h/2}^{h/2} y^2 dy = bh^3/12$$

$$I_y = hb^3/12$$

$$J_C = I_x + I_y = bh(h^2+b^2)/12$$

2.6 Triangle MOI

Base axis-এর জন্য similar triangle দিয়ে strip width $b(h-y)/h$ নেওয়া হয়। Integration করলে base MOI পাওয়া যায়।

$$I_{\text{base}} = bh^3/12$$

Triangle centroid base থেকে $h/3$ দূরে। Parallel-axis theorem ব্যবহার করলে centroidal axis parallel to base-এর MOI:

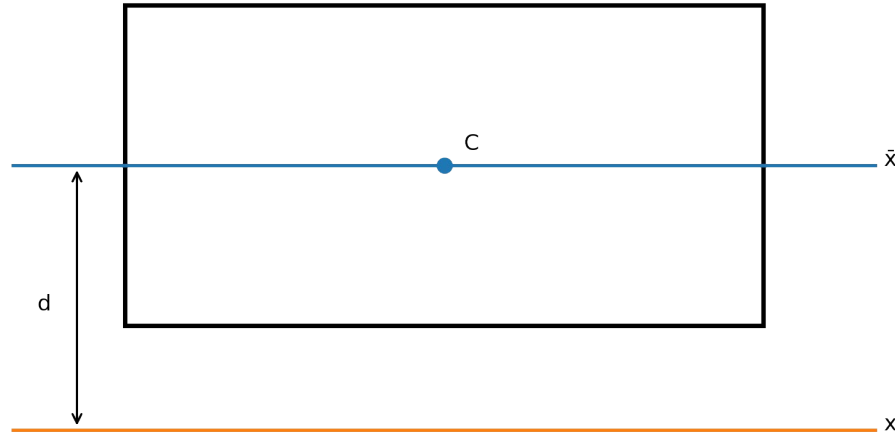
$$\bar{I}_x = I_{\text{base}} - A(h/3)^2 = bh^3/12 - (bh/2)(h^2/9) = bh^3/36$$

QB MATCH Matching Question Bank - Triangle MOI

Year(s)	Matching question / pattern	Priority
2023	Find moment of inertia and radius of gyration of a triangular area about a base.	★★★★★
Lecture slide	Derive rectangle and triangle area MOI results.	Derivation

2.7 Parallel-Axis / Transfer Formula

$$I = \bar{I} + A d^2$$



Given axis is parallel to centroidal axis and separated by distance d .

Centroidal axis থেকে d distance-এ parallel axis-এর MOI:

$$I = \bar{I} + A d^2$$

Proof: $(d-x)^2$ expand করলে cross term-এ $\int x \, dA$ আসে, centroidal axis-এর জন্য সেটি zero। ফলে $\bar{I} + Ad^2$ থাকে।

$$I = \int (d-x)^2 dA = \int x^2 dA - 2d \int x \, dA + d^2 \int dA = \bar{I} + Ad^2$$

$$\text{Radius of gyration form: } k^2 = \bar{k}^2 + d^2$$

QB MATCH Matching Question Bank - Parallel axis theorem

Year(s)	Matching question / pattern	Priority
2015	Prove $I = \bar{I} + Ad^2$; then use on composite unshaded area.	★★★★★
2016	Prove radius-of-gyration transfer form $k^2 = \bar{k}^2 + d^2$.	★★★★★
2017-2025	Composite MOI about parallel/centroidal axes.	★★★★★

2.8 Composite Area MOI - Procedure

- Composite shape-এর centroid আগে বের করুন (যদি centroidal axis required হয়)।
- প্রতিটি component-এর own centroidal MOI $\bar{I}_i$ লিখুন।
- Global axis থেকে component centroid-এর distance d_i নিন।
- $I_i = \bar{I}_i + A_i d_i^2$ ব্যবহার করুন।
- Hole/cutout হলে পুরো I_i term subtract করুন।
- সব component sum করে total I_x, I_y এবং দরকার হলে $J = I_x + I_y$ ।

Component	A_i	$\bar{I}_{xi}$	d_y	$I_{xi} = \bar{I}_{xi} + A_i d_y^2$
1	$+A_1$	$\bar{I}_1$	d_1	$+ (...)$
2	$+A_2$	$\bar{I}_2$	d_2	$+ (...)$
Hole	$-A_3$	$\bar{I}_3$	d_3	$- (\bar{I}_3 + A_3 d_3^2)$

Most common mistake: MOI formula-তে d হলো global axis থেকে component centroid-এর perpendicular distance; shape edge থেকে distance নয়।

QB MATCH Matching Question Bank - Composite Area MOI

Year(s)	Matching question / pattern	Priority
2015-2021	Compute I_x/I_y , centroidal/parallel MOI, polar MOI of shaded/unshaded sections.	★★★★★
2023	Composite cross-section with circular holes: polar MOI + radius of gyration.	★★★★★
2024	Composite area with semicircular cutouts: I_x and I_y .	★★★★★
2025	Centroidal MOI / polar MOI + radius of gyration for composite shapes.	★★★★★

2.9 A2 Formula Sheet

Quantity	Formula
First moment	$Q_x = A\bar{y}, Q_y = A\bar{x}$
Area MOI	$I_x = \int y^2 dA, I_y = \int x^2 dA$
Polar MOI	$J = \int r^2 dA = I_x + I_y$
Radius of gyration	$k = \sqrt{I/A}$
Parallel axis	$I = \bar{I} + Ad^2$
Polar transfer	$J = \bar{J} + Ad^2$ (parallel polar axes)

A2 one-line mnemonic: "Centroid first $\rightarrow$ own $\bar{I} \rightarrow$ shift $Ad^2 \rightarrow$ add/subtract."

SET A3 - Product of Inertia, Mass MOI & Pappus-Guldinus

এই set-এ তিনটি related but distinct idea আছে: (i) area-এর product of inertia, (ii) mass distribution-এর rotational inertia, (iii) centroid ব্যবহার করে surface/volume of revolution নির্ণয়ের Pappus theorem।

3.1 Product of Inertia of Area

$$P_{xy} = \int x y \, dA$$

P_{xy} area element-এর x এবং y - দুই coordinate একসাথে involve করে। তাই quadrant sign গুরুত্বপূর্ণ।



$$\text{I: } xy + \quad \text{II: } xy -$$

$$\text{III: } xy + \quad \text{IV: } xy -$$

I & III quadrants give xy positive; II & IV give xy negative.

Symmetry rule: যদি x বা y axis area-এর symmetry axis হয়, সেই centroidal axes-এর product of inertia zero হয়।

3.1.1 Transfer Formula

$$P_{xy} = \bar{P}_{xy} + A d_x d_y$$

d_x ও d_y centroidal axes থেকে shifted axes-এর signed distances। Product-এ sign হারাবেন না; কারণ $d_x d_y$ positive বা negative হতে পারে।

3.1.2 Maximum and Minimum Moment of Inertia

$$I_{\text{avg}} = (I_x + I_y)/2$$

$$R = \sqrt{\{(I_x - I_y)/2\}^2 + P_{xy}^2}$$

$$I_{\text{max}} = I_{\text{avg}} + R \quad ; \quad I_{\text{min}} = I_{\text{avg}} - R$$

$$k_{\text{max}} = \sqrt{I_{\text{max}}/A} \quad ; \quad k_{\text{min}} = \sqrt{I_{\text{min}}/A}$$

Correction note: Radius of gyration-এর unit length (in, mm, m) - length⁴ নয়। $k = \sqrt{I/A}$ ব্যবহার করলে unit নিজে যাচাই করা যায়।

QB MATCH Matching Question Bank - Product of inertia

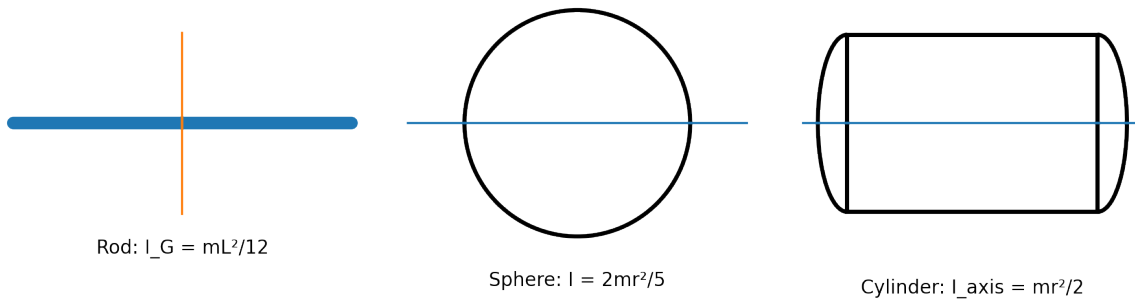
Year(s)	Matching question / pattern	Priority
2017-2022	Z-section product of inertia; often maximum/minimum I and radius of gyration.	★★★★★
2015	Define product of inertia + parallel-axis relation alongside area MOI.	★★★★☆
2023	Given I_x and I_y for angle section: maximum/minimum centroidal MOI, radius of gyration, product of inertia.	★★★★★
2025	Define product of inertia and explain structural-engineering applications.	★★★★★

3.1.3 Z-section Solving Pattern

14. Shape-কে rectangles-এ split করুন।
15. Centroidal x-y axes fix করুন।
16. Each rectangle-এর own centroidal $\bar{P}_{xy} = 0$ if axes parallel to its symmetry axes।
17. $P_{xyi} = A_i d_x d_y$ ব্যবহার করুন; quadrant sign keep করুন।
18. $P_{xy} = \sum P_{xyi}$; then I_x/I_y separately compute করে I_{max}/I_{min} বের করুন।

3.2 Moment of Inertia of Masses

Area MOI	Mass MOI
$I = \int r^2 dA$	$I = \int r^2 dm$
Geometric property	Mass distribution property
Unit: length ⁴	SI: kg·m ² ; FPS: slug·ft ²
Transfer: $I = \bar{I} + Ad^2$	Transfer: $I = \bar{I} + md^2$



Common mass-MOI formulas used in the supplied notes.

Body / axis	Mass MOI
Slender rod, transverse through center	$I_G = mL^2/12$
Slender rod, transverse through end	$I_{end} = mL^2/3$
Solid sphere about any diameter	$I = 2mr^2/5$
Solid cylinder about longitudinal axis	$I = mr^2/2$
Solid cylinder about transverse centroidal axis	$I = m(3r^2 + L^2)/12$
Solid cone about symmetry axis	$I = 3mr^2/10$

FPS unit shortcut: Weight W lb দেওয়া থাকলে mass $m = W/g$ slug, সাধারণত $g \approx 32.2 \text{ ft/s}^2$ । Formula-তে mass বসবে, weight নয়।

QB MATCH Matching Question Bank - Mass MOI

Year(s)	Matching question / pattern	Priority
2015	Mass MOI of cast-iron frustum of cone with cylindrical hole.	★★★★★
2023	Disk + shaft angular slowdown problem; rotational inertia enters dynamics.	★★★★☆
2024	Derive mass MOI of homogeneous right circular cylinder; frustum-type calculation.	★★★★★
2025	Derive sphere MOI; mallet composite mass MOI/radius of gyration.	★★★★★

3.2.1 Composite Mass MOI - Mallet Pattern

- Weight ~~mass~~ : $m = W/g$
- Handle-কে slender rod এবং head-কে cylinder হিসেবে model করুন।
- Requested axis each part-এর centroid দিয়ে না গেলে $I = \bar{I} + md^2$ ব্যবহার করুন।
- Total $I = \sum I_i$ এবং total mass $m = \sum m_i$
- Radius of gyration $k = \sqrt{I/m}$

3.2.2 Frustum / Hole by Subtraction

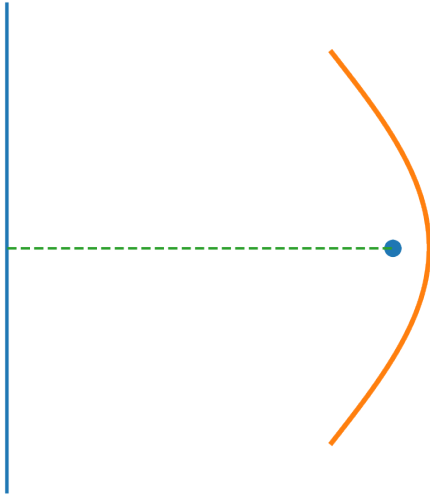
Frustum বা coaxial hole problem-এ “big solid – removed solid(s)” method সবচেয়ে পরিষ্কার। Similar triangle দিয়ে imaginary full cone height বের করে big cone minus small cone minus cylindrical hole করুন।

$$I_{net} = I_{big \text{ cone}} - I_{small \text{ cone}} - I_{hole}$$

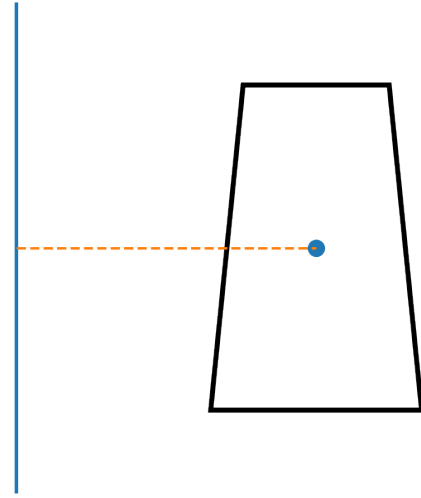
Sanity check: Removed material-এর inertia অবশ্যই subtract হবে। Resulting I positive এবং full solid-এর I -এর চেয়ে ছোট হওয়া উচিত।

3.3 Pappus-Guldinus Theorems

Theorem 1: $S = L \cdot 2\pi d$



Theorem 2: $V = A \cdot 2\pi d$



Centroid travels a circular path of length $2\pi d$ during revolution.

3.3.1 First Theorem - Surface Area of Revolution

একটি plane curve কোনো coplanar external axis-এর about ঘুরলে surface of revolution তৈরি হয়। Surface area = generating curve length $\times$ centroid যে distance travel করে।

$$S = L(2\pi d)$$

3.3.2 Second Theorem - Volume of Revolution

একটি plane area axis-এর about ঘুরলে volume of revolution = generating area $\times$ centroid travel distance।

$$V = A(2\pi d)$$

Use condition: Axis of revolution generating curve/area-এর plane-এ থাকে এবং সাধারণ application-এ generating region-এর interior cut করে না। d হলো centroid থেকে axis-এর perpendicular distance।

Generated body	Generating entity	Centroid distance	Pappus result
Cylinder surface	Line of length h at distance r	r	$S = 2\pi rh$
Cone lateral surface	Slant line L	$r/2$	$S = \pi rL$
Sphere surface	Semicircular arc length πr	$2r/\pi$	$S = 4\pi r^2$
Cone volume	Right triangle area $rh/2$	$r/3$	$V = \pi r^2 h/3$
Torus volume	Circle area πa^2 at distance b	b	$V = 2\pi^2 a^2 b$

QB MATCH Matching Question Bank - Pappus-Guldinus

Year(s)	Matching question / pattern	Priority
2015	State/use Pappus-Guldinus; paint inside/outside cone surface.	★★★★★

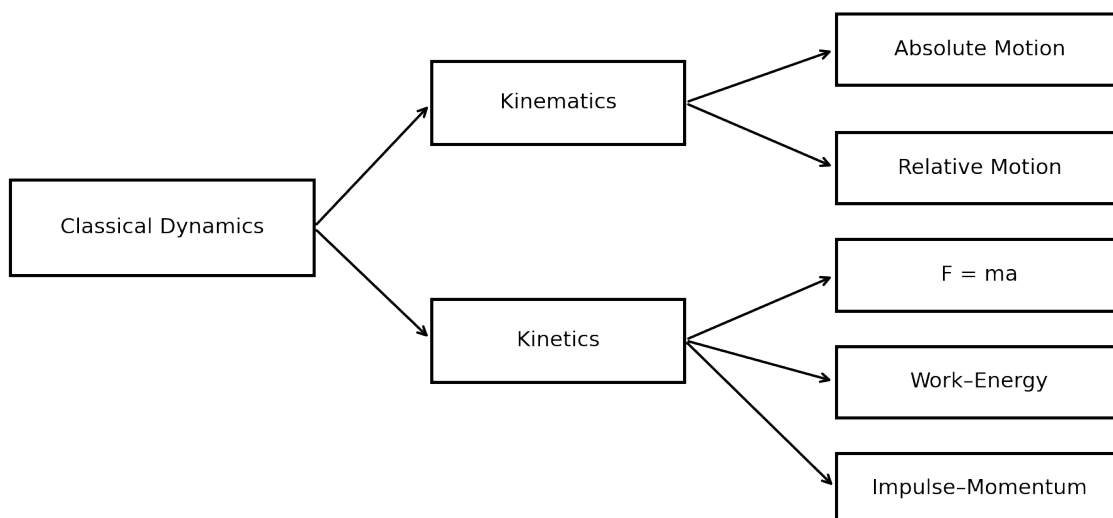
Year(s)	Matching question / pattern	Priority
2018	State area and volume theorems; calculate pulley rim volume/mass.	★★★★★
2023	Pulley rim mass/weight using Pappus-Guldinus theorem.	★★★★★
Lecture slides	Cone, sphere, torus; surface and volume examples.	Core

3.3.3 Pulley Rim Method

24. Cross-section-কে positive outer rectangle এবং negative inner cutout হিসেবে split করুন।
25. প্রতিটি area A_i এবং revolution axis থেকে centroid distance d_i বের করুন।
26. $V_i = A_i(2\pi d_i)$; cutout negative।
27. V_{total} sum করুন; mass $m = \rho V$ এবং weight $W = mg$ ।

A3 one-line mnemonic: Product: $\int xy \, dA$ • Mass MOI: $\int r^2 dm$ • Pappus: “measure $\times$ centroid travel distance”.

SET A4 - Dynamics, Work-Energy & Impulse-Momentum



Classical dynamics structure used in the supplied Section-A note.

Big picture: Kinematics বলে “কীভাবে motion হচ্ছে”; Kinetics বলে “কেন motion হচ্ছে / force-এর effect কী”। Kinetics-এ তিনটি main route: $F=ma$, Work-Energy, Impulse-Momentum।

4.1 Classical Dynamics: Kinematics vs Kinetics

Branch	Concern	Typical variables	Best method when...
Kinematics	Geometry/time of motion	s, v, a, t	Force not required
Kinetics - $F=ma$	Force ↔ acceleration	F, m, a	Acceleration/time relation needed
Work-Energy	Force through displacement	U, v, s	Velocity-position; no time needed
Impulse-Momentum	Force through time	J, v, t	Velocity-time / impact / jet

Method selection shortcut: Time নেই কিন্তু displacement আছে → Work-Energy. Time interval/impact আছে → Impulse-Momentum. Acceleration বা tension directly চাই → $F=ma$ ।

4.2 Constant-Acceleration Kinematics

$$v = u + at$$

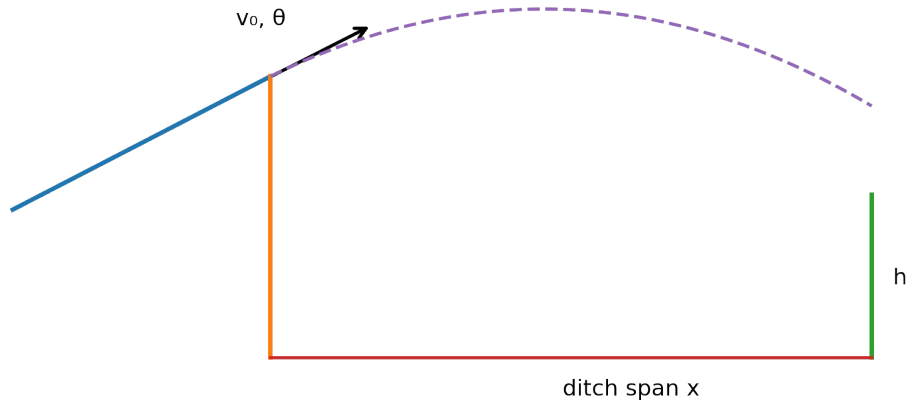
$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

Sign convention আগে fix করুন। Negative a মানে chosen positive direction-এর বিপরীতে acceleration; system “reverse” হতে পারে।

4.3 Plane / Projectile Motion

Body-এর সব particle একই plane-এ move করলে plane motion বলা হয়। Motorcycle ditch problem-এ rider-কে particle হিসেবে model করে projectile equation ব্যবহার করা হয়।



Projectile model for the recurring motorcycle ditch problem.

$$x = v_0 \cos\theta \cdot t$$

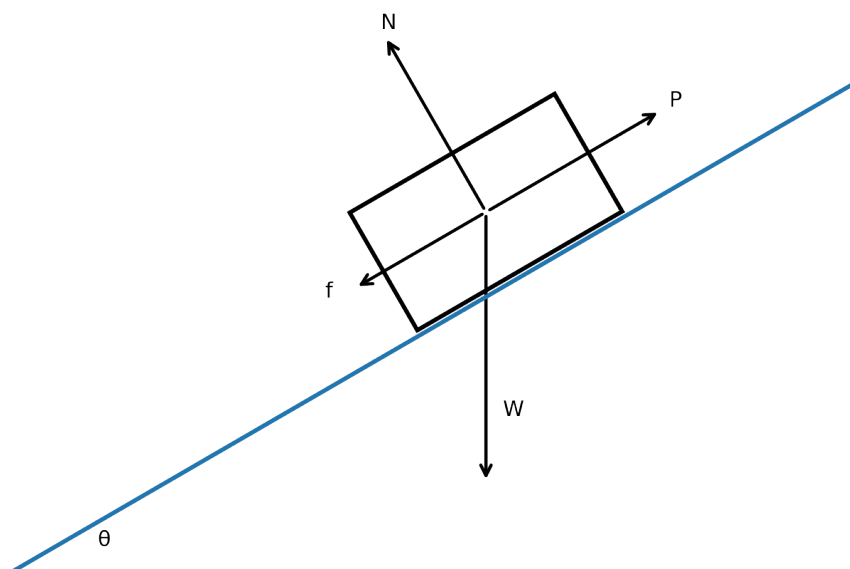
$$y = v_0 \sin\theta \cdot t - \frac{1}{2}gt^2$$

$$\text{Eliminating } t: y = x \tan\theta - \frac{gx^2}{2v_0^2 \cos^2\theta}$$

QB MATCH Matching Question Bank - Plane/projectile motion

Year(s)	Matching question / pattern	Priority
2015, 2018, 2021	Motorcycle rider jumps a 20-ft ditch; find maximum h.	★★★★★
Dynamics note	Same projectile pattern with $v_0 \approx 70$ mph and $\theta \approx 30^\circ$.	Core practice

4.4 Force-Mass Acceleration Method



Typical inclined-block FBD used before writing $\Sigma F = ma$.

28. Body আলাদা করে FBD আঁকুন।
29. Normal direction এবং motion direction axis নিলে equations সহজ হয়।
30. Friction $f = \mu N$, direction motion/impending motion-এর opposite।
31. $\Sigma F_{\text{parallel}} = ma$ এবং $\Sigma F_{\text{normal}} = 0$ (if no normal acceleration)।
32. Pulley/cable constraints থাকলে connected bodies-এর acceleration relation লিখুন।

FPS reminder: Equation $\Sigma F = ma$ -তে $m = W/g$ । Weight lb-কে সরাসরি mass হিসেবে বসাবেন না।

QB MATCH Matching Question Bank - $F=ma$ dynamics

Year(s)	Matching question / pattern	Priority
2022 note pattern	Blocks on incline with different μ ; elapsed time until contact.	★★★★☆
Class note	Block + movable pulley: compute accelerations and cable tension.	★★★★☆
2024/2025 recent	Accelerating cart / overturning box / pulley-linked body dynamics.	★★★★☆

4.5 Work

Work হলো force-এর displacement direction component $\times$ displacement-এর accumulation। Constant force হলে:

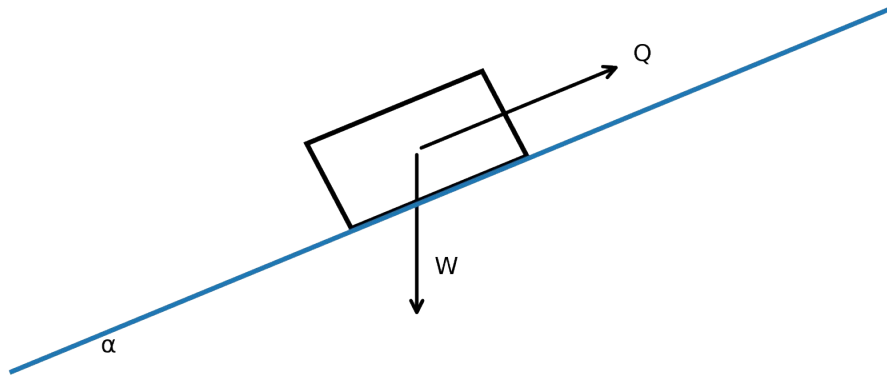
$$U = F s \cos\phi$$

Force	Work sign
Component along motion	Positive
Opposes motion (friction)	Negative
Normal reaction perpendicular to motion	Zero
Gravity	Sign depends on up/down displacement
Spring	Can be positive or negative depending compression/extension and motion

Net work: একটি rigid body-তে বিভিন্ন force-এর individual work algebraically যোগ করলে net work পাওয়া যায়।

4.6 Kinetic Energy & Work-Energy Principle

$$\Sigma U = T_2 - T_1$$



Work-energy is especially useful when displacement and speed are involved.

$$\text{Translational KE: } T = \frac{1}{2}mv^2$$

$$\text{Rotational KE: } T_{\text{rot}} = \frac{1}{2}I\omega^2$$

$$\text{Work-Energy: } \Sigma U_{1 \rightarrow 2} = T_2 - T_1 = \Delta T$$

4.6.1 Short Derivation

Newton-এর law $R = ma$ এবং $a = v(dv/ds)$ ব্যবহার করুন।

$$R ds = m v dv$$

$$\int R ds = \int m v dv = \frac{1}{2}m(v_2^2 - v_1^2)$$

$$\text{Therefore: } W_{\text{net}} = KE_2 - KE_1 = \Delta KE$$

QB MATCH Matching Question Bank - Work-energy proof & application

Year(s)	Matching question / pattern	Priority
2016	Define work and power; prove $W_{\text{net}} = \Delta KE$; chimney work against gravity.	★★★★★
2018	Explain work & kinetic energy; inclined body net work; impulse proof; chimney work.	★★★★★
2023	What is classical dynamics? Prove $W_{\text{net}} = \Delta KE$; incline resultant/net work; masonry chimney.	★★★★★
2025	Ore car speed change under drawbar pull/resistance - work-energy.	★★★★★

4.7 Work Against Gravity - Chimney Pattern

Tall masonry chimney-তে প্রতিটি thin slice-কে ground থেকে y height-এ lift করতে $dU = (\text{weight of slice}) \cdot y$ লাগে। Constant cross-sectional area A এবং unit weight γ হলে:

$$dW = \gamma A dy \quad ; \quad dU = y \cdot dW = \gamma A y dy$$

$$U = \int_0^H \gamma A y dy = \gamma A H^2/2$$

If hollow circular: $A = (\pi/4)(D_o^2 - D_i^2)$ । Units consistent রাখুন।

4.8 Power

$$P = dU/dt = F \cdot v \quad (\text{for force parallel component})$$

Rotational case-এ $P = M\omega$ । Power হলো work করার rate।

4.9 Impulse and Linear Momentum

Impulse হলো force $\times$ time-এর accumulation। এটি momentum change তৈরি করে।

$$J = \int_{t_1}^{t_2} F dt$$

$$\int F dt = m v_2 - m v_1$$

4.9.1 Proof

$$F = m dv/dt \Rightarrow F dt = m dv$$

$$\int F dt = \int m dv = m(v_2 - v_1)$$

Constant force shortcut: F constant হলে $J = F\Delta t$ । Multiple force থাকলে vector resultant or component-wise impulse নিন।

QB MATCH Matching Question Bank - Linear impulse/momentum

Year(s)	Matching question / pattern	Priority
2015	State principle; water/steam jet force problem.	★★★★★
2016, 2017, 2018, 2019/2021 pattern	State/prove impulse-momentum; block/friction numerical.	★★★★★
2023	State principle of linear impulse and momentum; related dynamics terms.	★★★★★
2024	Prove $R\Delta t = \Delta(mv)$; impact/velocity problem.	★★★★★
2025	Jet on fixed blade + direct central impact problem.	★★★★★

4.10 Angular Impulse and Momentum

Fixed-axis rotation-এ moment/torque-এর time integral angular momentum change-এর সমান।

$$\int M_O dt = I_O(\omega_2 - \omega_1)$$

When useful: Disk/shaft slowing down, bearing friction torque, rotating pulley/disk problems।

4.11 Fluid Jet / Blade Pattern

Jet problem-এ momentum equation component-wise apply করুন। Inlet/outlet velocity direction carefully resolve করতে হয়। Steady flow-এর জন্য force component mass-flow rate $\times$ velocity change-এর সঙ্গে related।

$$\Sigma F_x = \dot{m}(V_{2x} - V_{1x}), \quad \Sigma F_y = \dot{m}(V_{2y} - V_{1y})$$

FPS note: যদি flow rate weight/time হিসেবে দেওয়া হয়, mass-flow $\dot{m} = W/g$ ব্যবহার করুন।

4.12 Direct Central Impact - Recent Pattern

Impact-এ external impulse negligible হলে system linear momentum conserved। Coefficient of restitution e relative separation speed / relative approach speed।

$$m_A u_A + m_B u_B = m_A v_A + m_B v_B$$

$$e = (v_B - v_A)/(u_A - u_B) \quad (1D \text{ sign convention consistent রাখতে হবে})$$

Sign discipline: একটি direction positive fix করে সব velocity signed quantity হিসেবে লিখুন। Magnitude দিয়ে equation লিখলে impact problem-এ sign error খুব হয়।

QB MATCH Matching Question Bank - Impact

Year(s)	Matching question / pattern	Priority
2025	Two freight cars in direct central impact: post-impact velocities, impulse, loss of kinetic energy.	★★★★★

4.13 A4 Problem-Selection Guide

Question clue	Use first
"Find acceleration / tension / elapsed time"	$F = ma$ + kinematics
"Find speed after moving s"	Work-Energy
"Force acts for t sec / impact / jet"	Impulse-Momentum
"Maximum h across ditch"	Projectile equation
"Rotating disk slows from rpm_1 to rpm_2 "	Angular impulse-momentum
"Work against gravity in chimney"	Integration of $\gamma A y \, dy$

A4 one-line mnemonic: Acceleration → Force; Distance → Energy; Time/Impact → Impulse.

Section A - Topic-to-Question-Bank Master Map

এই table-টি revision-এর সময় দ্রুত matching করতে ব্যবহার করুন। Priority repetition + recency দুটো বিবেচনা করে সাজানো।

Priority	Topic	Regular QB evidence / recurring wording
★★★★★	Composite centroid	2015-2021 repeatedly; 2023 unshaded area; 2024 composite/parabolic; 2025 composite C.G.
★★★★★	Centroid theory + sector	Definition/applications/axis symmetry/triangle/sector repeatedly; 2023 & 2025 sector
★★★★★	Composite area MOI	2015-2021 recurring; 2023 holes + polar/k; 2024-25 centroidal/polar MOI
★★★★★	$J = I_x + I_y$ / radius / parallel axis	Repeated theory/proof and then numerical application
★★★★★	Impulse-momentum	Repeated proof + block/jet; 2025 impact/jet
★★★★★	Work-energy	2016/2018/2023 proofs/applications; 2025 ore-car work-energy
★★★★★	Product of inertia / principal I	Z-section repeated 2017-2022; 2023 angle section; 2025 applications
★★★★☆	Mass MOI	Frustum, cylinder/sphere derivations, mallet, rotating disk
★★★★☆	Pappus-Guldinus	2015 cone paint; 2018 & 2023 pulley rim
★★★★☆	Projectile motorcycle	2015/2018/2021 family

Most Important Theory Questions

- Define centroid; distinguish centroid, center of mass and center of gravity.
- Explain real-life applications of centroid in building/structural engineering.
- Describe axis of symmetry / principle of symmetry.
- Define rectangular MOI, polar MOI, radius of gyration and product of inertia.
- Explain significance/applications of first and second moments of area.
- State Pappus-Guldinus theorems for surface area and volume.
- Define classical dynamics; distinguish kinematics and kinetics.
- Define work, kinetic energy, power, impulse and momentum.
- State principle of linear impulse and momentum.

Most Important Proof / Derivation List

Derivation	Priority
Centroid of triangle: $\bar{y} = h/3$ from base	★★★★★
Sector/arc centroid formula	★★★★☆

Derivation	Priority
$J = I_x + I_y$	★★★★★
Rectangle MOI about centroidal axes	★★★★☆
Triangle MOI about base / centroidal parallel axis	★★★★☆
Parallel-axis theorem $I = \bar{I} + Ad^2$	★★★★★
Radius transfer $k^2 = \bar{k}^2 + d^2$	★★★★☆
Pappus theorem statement + application logic	★★★★★
Work-energy: $W_{\text{net}} = \Delta KE$	★★★★★
Linear impulse-momentum: $\int F dt = \Delta(mv)$	★★★★★
Mass MOI of cylinder/sphere (recent trend)	★★★★☆

3-out-of-4 Set Strategy

যেহেতু Section A-তে ৪ set-এর মধ্যে ৩টি answer করতে হবে, preparation এমন হওয়া উচিত যাতে exam hall-এ choice থাকে।

Plan	Recommended preparation
Safe plan	A1 + A2 + A4 strong; A3 backup basics + repeated numericals
Math-strong plan	A1 + A2 + A3 strong; A4 at least work-energy & impulse
Dynamics-strong plan	A1 + A4 strong; A2 standard formulas/composite; A3 Pappus/product selected
Never do	Exactly 3 set only, no backup - one unusual question can remove your choice

Final revision rule: প্রতিটি set-এর জন্য ১টি theory answer + ১টি derivation + ২টি representative numerical pattern লিখে practice করুন। শুধু পড়লে mechanics exam-এ speed তৈরি হয় না।

Source Basis & Note Scope

Prepared from the class notes and lecture slides supplied in this conversation, including the centroid, area moment of inertia, product of inertia, mass moment of inertia, Pappus-Guldinus, dynamics, work-energy and impulse-momentum materials, and matched against the supplied regular question bank (2015-2021, 2023, 2024-25). Backlog examination is intentionally not used as the main regular-exam matching base in this Section-A note.

Some source scans contain typographical/notation inconsistencies. Where the same concept is clearly supported by multiple supplied sources, this note presents a consistent standard notation (e.g., $k = \sqrt{I/A}$, units of radius of gyration as length).

Made by 2223033